

GEN AI

ASSESSMENT – 1

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Subject: Generative AI

1. What is the advantage of using {user_input} instead of hard-coding input?

Answer: Using {user_input} makes the prompt dynamic and reusable. Different users can provide different inputs without changing the prompt or code.

- It separates the prompt logic from the actual data, making the system more maintainable.
- It allows the same prompt template to be used across multiple applications or use cases.
- It reduces the risk of errors since the code does not need to be edited for every new input.
- It supports automation, as inputs can be supplied programmatically from other systems.

2. When would .format() be enough?

Answer: .format() is enough when we only need to insert simple variables into a prompt. It is easy and suitable for basic prompt templates.

- It works well for small-scale projects or quick prototypes with minimal logic.
- It is efficient when there is no need for conditional statements or loops in the prompt.
- It keeps the code simple and readable when only a few placeholders need to be filled.
- It is a good choice when performance and simplicity matter more than flexibility.

3. When might Jinja2 be useful?

Answer: Jinja2 is useful for complex and dynamic prompts. It supports conditions, loops, and reusable templates.

- It is helpful when prompts need to change based on multiple variables or user roles.
- It allows the use of loops to generate repeated sections, such as lists or multi-turn conversations.
- It supports template inheritance, making it easier to maintain large sets of related prompts.
- It improves scalability when building prompt pipelines for production-level applications.

4. Why should prompts be versioned?

Answer: Prompts should be versioned to track changes and compare results. If a new version performs worse, we can return to the previous version.

- Versioning helps maintain a clear history of how prompts have evolved over time.

- It enables collaboration, as multiple team members can experiment without overwriting each other's work.
- It supports A/B testing to identify which prompt version yields better outcomes.
- It improves accountability and reproducibility of results in AI-based systems.

5. What is a prompt regression?

Answer: A prompt regression occurs when a new prompt version performs worse than the previous version.

- It can happen due to unintended side effects when modifying wording or structure.
- Regressions may reduce accuracy, relevance, or consistency of the model's responses.
- Detecting regressions early helps prevent poor-quality outputs from reaching production.
- Regular evaluation and monitoring are needed to catch regressions before deployment.

6. Why use the same evaluation set when comparing prompt versions?

Answer: Using the same evaluation set makes the comparison fair and consistent. It helps us determine whether a performance change is caused by the prompt.

- It eliminates variability caused by differing test data, ensuring a controlled comparison.
- It allows accurate benchmarking of improvements or regressions across versions.
- It builds confidence in decision-making when choosing the best-performing prompt.
- It supports reproducibility, so results can be verified and trusted by others.